

VAIT: Visuotactile-Guided Adaptive In-hand Twisting of Slender Objects

Abstract—The manipulation of slender objects poses inherent challenges due to their compliance, low torsional stiffness, and high sensitivity to surface friction. Existing robotic solutions face a trade-off between precision and versatility: specialized grippers achieve submillimeter accuracy but are rigid by design, failing in dynamic or multi-object workflows; general-purpose grippers provide adaptability but lack stable contact and fine force control essential for delicate tasks. This gap leaves many dexterous, multistage manipulations beyond the reach of current automation. In this work, we introduce VAIT (Visuotactile-guided Adaptive In-hand Twisting), a control framework designed for robust slender object manipulation. VAIT leverages a visuotactile-sensorized, multi-degree-of-freedom (DoF) gripper to achieve continuous force–position coupled in-hand twisting operations. With a single end effector, the system robustly handles objects ranging from submillimeter guidewires to 5 mm rods, without requiring task-specific hardware modifications. Experimental results demonstrate that VAIT effectively combines precision and versatility, enabling the automation of complex multi-object and multistage manipulation tasks previously inaccessible to robots.

I. INTRODUCTION

The dexterous manipulation of slender objects remains a core challenge in robotics. These objects exhibit high compliance, low torsional stiffness, and strong sensitivity to surface friction, making grasp stability and in-hand control particularly difficult. Such challenges are ubiquitous in industrial applications, including cable assembly [1] and optical fiber-related workflows [2], as well as in medical scenarios such as catheter and guidewire insertion [3].

Existing gripper designs generally fall into two extremes. On one hand, specialized grippers can achieve high positional accuracy through rigid kinematic constraints [4]. Although effective in narrowly defined domains, these devices often lack the versatility to accommodate diverse shapes, sizes, or dynamic multistage workflows. On the other hand, general-purpose grippers provide adaptability but often fail to maintain stable contact or deliver fine force control during delicate interactions [5]. This trade-off greatly complicates the automation of complex tasks, where precision and adaptability are simultaneously required.

A promising direction to overcome this limitation lies in visuotactile sensing, which integrates vision and tactile deformation cues into a unified modality. By embedding cameras and markers into soft elastomeric layers, such sensors provide rich real-time feedback on contact geometry, force distribution, and incipient slip [6]. Despite these advances, a general framework for continuous force-position coupled manipulation of slender objects remains largely unexplored. In particular, rolling or twisting control, which

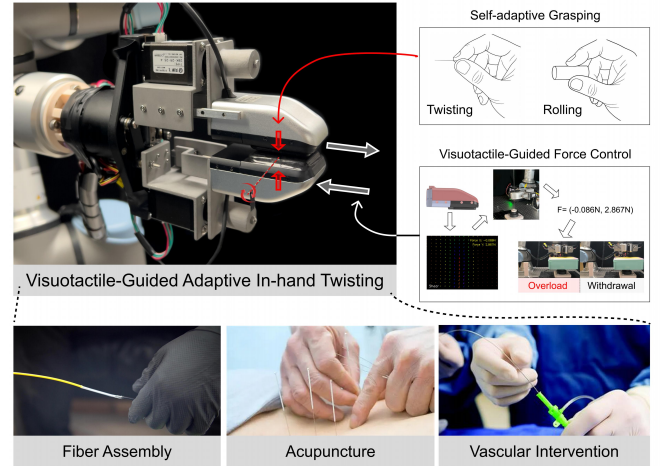


Fig. 1. Overview of the visuotactile-guided adaptive in-hand twisting (VAIT) robot. A multi-DoF visuotactile gripper provides self-adaptive grasping and force–position coupled control for robust rolling and twisting of slender objects while maintaining stable contact, with potential applications to fiber assembly, acupuncture, and vascular intervention.

requires simultaneously regulating forces and kinematics while maintaining stable contact, has received little attention [7].

To address this gap, we propose Visuotactile-guided Adaptive In-hand Twisting (VAIT), a control framework designed for robust slender object manipulation. As illustrated in Fig. 1, VAIT enables self-adaptive grasping and visuotactile-guided force control, which are essential for tasks such as fiber assembly, acupuncture, and vascular intervention. Specifically, this work makes three key contributions.

- We build a visuotactile-based, multi-DoF robot for grasping and twisting extremely slender objects. By exploiting the elasticity of the silicone skins on the visuotactile pads, the system provides stable grasping forces exceeding 0.74 N on filaments as thin as 0.2 mm.
- We develop the VAIT model to realize rolling-angle control and force measurement on submillimeter filaments. Through simulation analysis, creep induced by silicone compression is compensated, keeping the maximum rolling-angle error within 4° .
- We also design two filament twisting experiments and conduct a systematic evaluation of the proposed robot. By enforcing preset safe operation forces, we demonstrate the practical applicability of our method in real-world scenarios.

The remainder of this paper is organized as follows. Section II reviews related work on slender object manipulation.

Section III presents the design of our robot structure. Section IV introduces the VAIT framework, including its modeling scope, kinematic rolling regime, frictional sliding regime, and unified control model. Section V provides simulation analyses, while Section VI details experimental validations. Finally, Section VII discusses limitations, potential applications, and future directions.

II. RELATED WORK

The achievement of dexterous manipulation in the hand remains a central goal in robotics, with the continuous axial rotation or twisting of an object being a key benchmark of this skill [8], [9]. Research towards this primitive has largely bifurcated: one path uses complex, model-based algorithms like trajectory optimization on general-purpose dexterous hands to achieve submillimeter precision on large, rigid objects [10]; the other path involves designing specialized hardware with active surfaces, such as belts or deployable springs, to mechanically simplify the control of twisting for palm-sized objects, often prioritizing the novel mechanism over ultimate precision [11], [12]. While effective in their respective domains, these approaches have predominantly focused on objects where establishing a stable grasp is straightforward.

The challenge of in-hand twisting is significantly amplified when applied to Deformable Linear Objects (DLOs), particularly the subclass of ultra-thin slender objects such as fibers and catheters, where maintaining a stable grasp is non-trivial. The DLO manipulation literature has largely focused on higher-level tasks like cable routing [13] or on fundamental primitives like linear following, where a gripper slides along the object's length [14]. When axial rotation is required in these contexts, it is typically accomplished through coarse and inefficient reorientation of the entire robot arm, or through passive and sensorless mechanisms that lack adaptability for objects with varying compliance [15]. This reveals a lack of a robust, fundamental primitive for precise, in-hand twisting tailored to the unique challenges of slender DLOs.

Overcoming these challenges requires rich perceptual feedback at the point of contact, a capability offered by modern visuotactile sensors [16]. While traditional tactile sensing often focuses on regulating normal forces for grasp stability, these camera-based sensors can also measure high-resolution shear forces by tracking the lateral displacement of internal markers. This shear information provides a direct measurement of the tangential forces that generate rotational torque, a capability that is largely absent in traditional tactile arrays. Although prior work with dexterous hands has begun to leverage visuotactile feedback for DLO manipulation, the focus has remained on using contact localization and normal forces for linear sliding [14]. The critical shear data has not yet been fully exploited for adaptive, closed-loop control of twisting. This work addresses this gap by presenting VAIT, a framework that leverages shear-based visuotactile feedback to enable high-precision, adaptive in-hand twisting of ultra-thin slender objects.

III. VISUOTACTILE-BASED GRIPPER

A. Gripper Structure

VAIT robotic system is equipped with a visuotactile-based gripper that integrates compliant sensing with multi-DoF actuation for the manipulation of slender objects. As illustrated in Fig. 2(a), the gripper consists of two opposing visuotactile sensors (DW-Tac W, Daimon, China, denoted as VS1 and VS2 in Fig. 2(b)) mounted on parallel linear guides (TGN5C, China, denoted as LG1 and LG2 in Fig. 2(b)), each driven by an independent linear motor (28HS32, China, denoted as LM1 and LM2 in Fig. 2(b)). In addition, a third linear guide (MGN20, China, denoted as LG3 in Fig. 2(b)) in combination with a gripper motor (DM-J4310, Damiao, China, denoted as GM in Fig. 2(b)) provides the additional degree of freedom required for grasping. This configuration enables precise bidirectional translation of the visuotactile sensors while simultaneously allowing controlled rotational manipulation of the object.

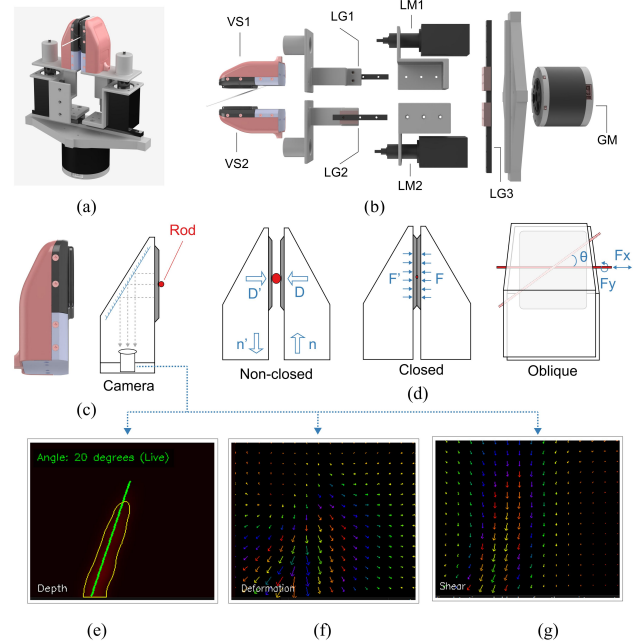


Fig. 2. **Design of the visuotactile-based gripper.** (a) Overall view of the gripper structure. (b) Assembly showing linear motors and actuation components. (c) Visuotactile sensor module. (d) Example of slender-object grasping. (e–g) Visuotactile sensory outputs, including (e) Depth maps, (f) Deformation maps, and (g) Shear stress maps.

B. Angle and Force Estimation

Since slender objects can be grasped at arbitrary orientations, we estimate their deviation angle with respect to the vertical axis from visuotactile depth maps (Fig. 2(e)). The centerline is extracted and its orientation is computed from the endpoints $(\Delta x, \Delta y)$:

$$\theta = \arctan 2(\Delta y, \Delta x) - \frac{\pi}{2}, \quad \theta^\circ = \theta \cdot \frac{180}{\pi}. \quad (1)$$

The angle is normalized to $[-90^\circ, 90^\circ]$ and expressed as an unsigned offset

$$\theta_{\text{raw}}^{\circ} = |\theta_{\text{norm}}^{\circ}|. \quad (2)$$

To reduce noise, a two-stage smoothing is applied. A rolling average over the latest n_t values is computed as

$$\bar{\theta}_t = \frac{1}{n_t} \sum_{i=1}^{n_t} \theta_i, \quad (3)$$

and then fused with the previous valid estimate θ_{t-1} using exponential smoothing:

$$\theta_t = \begin{cases} \alpha \theta_{t-1} + (1 - \alpha) \bar{\theta}_t, & \text{if } \theta_{t-1} \text{ exists,} \\ \bar{\theta}_t, & \text{otherwise.} \end{cases} \quad (4)$$

If detection fails, the system retains θ_{t-1} when the gripper is closed, or resets to null when open. This ensures reliable estimation of object-gripper misalignment under variable contact conditions. In addition, shear stress map (Fig. 2(g)) distributions are obtained from optical flow of surface markers, represented as arrow fields with magnitude m_i and direction vector $(d_{x,i}, d_{y,i})$ [6]. To account for arbitrary orientations, these vectors are rotated by the estimated angle θ_t , and the shear integrals S are defined as

$$S_x = \sum_{i=1}^N m_i (d_{x,i} \cos \theta_t + d_{y,i} \sin \theta_t), \quad (5)$$

$$S_y = \sum_{i=1}^N m_i (-d_{x,i} \sin \theta_t + d_{y,i} \cos \theta_t). \quad (6)$$

The external forces are then estimated through a linear mapping:

$$F_x = a_1 S_x + b_1, \quad F_y = a_2 S_y + b_2. \quad (7)$$

Due to the isotropy of the silicone material, the coefficients are equal,

$$a_1 = a_2 = a, \quad b_1 = b_2 = b. \quad (8)$$

C. Calibration

To determine the coefficients a and b in the linear mapping, a force calibration experiment was conducted. As illustrated in Fig. 3(a), the gripper grasped a slender filament and pulled it vertically upward by a prescribed displacement. During this process, the shear integral S was computed from the visuotactile sensor output, while the corresponding real force was simultaneously recorded by the force sensor placed beneath the specimen (MiniONE Pro, Bota, Switzerland). The relationship between S and the measured force F was then analyzed and fitted, as shown in Fig. 3(b). The calibration yielded a linear model with coefficients $a = 0.0053$ and $b = 1.0788$.

To further verify the consistency of the gripper's performance across objects of different diameters, additional measurements were conducted using steel rods of 1.0 mm, 0.5 mm, and 0.2 mm in diameter, as shown in Fig. 3(c). For each specimen, the maximum gripping force was recorded.

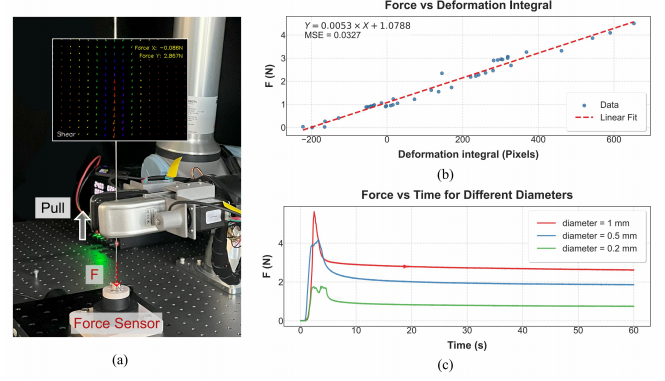


Fig. 3. **Force calibration.** (a) Setup of the force calibration experiment. (b) Linear fitting between shear integral S and measured force F . (c) Maximum gripping forces on steel rods with different diameters.

Due to the viscoelastic creep of the silicone pads, the sustained gripping force gradually decreased over time after reaching its peak. This creep behavior reflects the time-dependent deformation of silicone material under constant loading, which leads to a slow relaxation of the applied gripping force. Nevertheless, the gripper was able to maintain maximum forces of 2.62 N for the 1.0 mm rod, 1.84 N for the 0.5 mm rod, and 0.74 N for the 0.2 mm rod, demonstrating reliable force delivery across varying object scales.

IV. VISUOTACTILE-GUIDED ADAPTIVE IN-HAND TWISTING

A. Modeling Scope and Objective

The objective of our control algorithm is to establish a robust and predictable model for the local rotational displacement of a slender object at the point of gripper contact. In practical applications, the global behavior of a flexible object is complex; due to its compliance, a rotation applied locally does not produce a trivially predictable pose change at the object's distal tip.

Therefore, our work does not attempt to model the full-body dynamics of the manipulated object. Instead, we focus on a foundational prerequisite: a high-fidelity model of the local manipulation action itself. By precisely modeling and controlling the rotational displacement at the contact interface, we establish a reliable manipulation primitive. This primitive serves as an essential building block for higher-level planners, while the broader challenge of relating local rotation to global object pose is considered beyond the scope of this work.

B. Regime 1: Kinematic Rolling in Open Configuration

When the gripper fingers grasp an object without making contact with each other, the system operates in an **Open Configuration (OC)**. In this regime, the manipulation is primarily governed by kinematic rolling contact, which we model using an idealized rack-and-pinion analogy.

The ideal kinematic model assumes the two opposing fingers (the "racks") induce a pure rolling motion on the cylindrical object (the "pinion"). Given a linear travel per

motor step, s , the effective linear displacement imparted to the object's surface for n steps is $2sn$. The ideal angular displacement, Θ_{ideal} , in degrees is thus:

$$\Theta_{\text{ideal}} = \frac{720sn}{\pi d}. \quad (9)$$

To bridge the gap between this idealized model and the performance of our physical system, we introduce three compensation factors that account for key non-ideal effects:

- **Angular Misalignment** ($\cos(\theta)$): The ideal model assumes the actuation is perfectly normal to the object's primary axis. In practice, an object orientation, θ , reduces the effective torque, a geometric projection corrected by a $\cos(\theta)$ term.
- **Continuous Dynamic Losses** (K_{OC}): Ideal motion transfer is impeded by system dynamics. A linear gain factor, K_{OC} , accounts for proportional losses stemming from the compliance of the sensor material and kinetic friction at the contact interface.
- **Initial Motion Threshold** (C_{OC}): The system's elastomeric surfaces exhibit a "preloading" phenomenon. An initial number of motor steps is absorbed to generate sufficient strain to overcome static friction, creating a motion dead zone. This is modeled by a constant offset, C_{OC} .

Incorporating these factors yields our final, compensated OC model:

$$\Theta_{\text{OC}}(d, n, \theta) = K_{\text{OC}} \left(\frac{720sn}{\pi d} \cos(\theta) \right) + C_{\text{OC}}. \quad (10)$$

The empirical identification of the gain coefficient K_{OC} and the offset C_{OC} is detailed in Section IV-E.

C. Regime 2: Frictional Sliding in Closed Configuration

The transition into this regime depends on the applied compressive force, under which three discrete contact states can be distinguished. At incipient contact, the grip is insufficient, as the normal force cannot consistently overcome static friction, leading to intermittent slip and non-repeatable rotation. With increased compression, the system reaches a minimal viable grip where rotations can be executed with high repeatability, but the induced deformations remain below the detection threshold of the visuotactile sensors, thus precluding feedback. A further increase in force establishes a stable visuotactile grip in which manipulation becomes reliable and the corresponding shear forces yield a robust, measurable signal; however, the compression must be carefully controlled to permit sliding while avoiding motor overload, step loss, and loss of accuracy. In our framework, we formally define the **Closed Configuration (CC)**, as it uniquely guarantees both mechanical predictability and the rich sensory feedback required for a visuotactile-guided system.

While the OC model structure was initially considered as a baseline, our empirical and simulation analysis revealed a more complex relationship for the CC state. The resulting model incorporates an additional term to account for

diameter-dependent effects that are prominent in this high-contact state:

$$\Theta_{\text{CC}}(d, n) = K_{\text{CC}} \left(\frac{720sn}{\pi d} \right) + \frac{A_{\text{CC}}}{d} + C_{\text{CC}}. \quad (11)$$

This model deviates from the open configuration by incorporating a **Diameter-Dependent Correction** (A_{CC}/d). This empirical term captures second-order effects prominent in the closed state, such as how the compliant pads envelop the object, an effect that is negligible in the low-compression OC state but becomes significant under the high compressive forces of the CC state. As object diameter decreases (i.e., curvature increases), localized shear stresses and deformation patterns introduce a non-linear behavior that this term effectively models. The detailed justification for this model's form, along with the empirical and simulation-based identification of the coefficients K_{CC} , A_{CC} , and C_{CC} , is presented in Section V.

D. Unified Rotational Pose Model

The models for these two distinct physical regimes are synthesized into a single, unified piecewise control law. The ability to transition between these models based on the detected contact state is the foundation of our system's adaptive capability.

$$\Theta_{\text{deg}}(d, n, \theta) = \begin{cases} K_{\text{OC}} \left(\frac{720sn}{\pi d} \cos(\theta) \right) + C_{\text{OC}} & \text{OC,} \\ K_{\text{CC}} \left(\frac{720sn}{\pi d} \right) + \frac{A_{\text{CC}}}{d} + C_{\text{CC}} & \text{CC.} \end{cases} \quad (12)$$

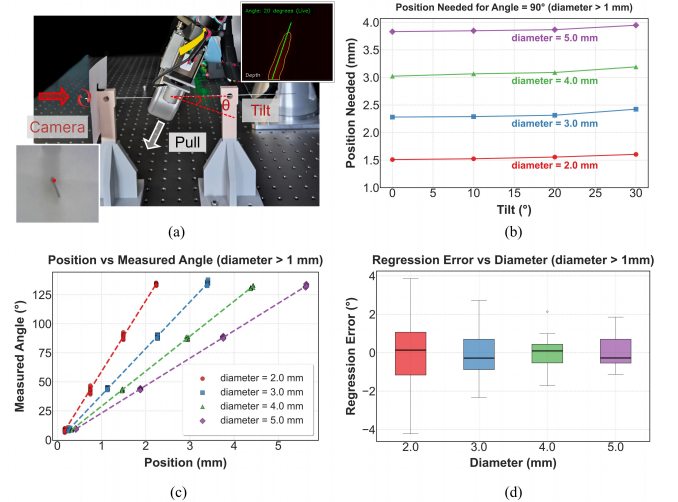


Fig. 4. **OC model calibration.** (a) Experimental setup for measuring rod rotation under different diameters and tilt angles. (b) Example results showing that achieving a constant 90° rotation requires greater displacement for larger diameters and tilt angles. (c) Linear regression between the kinematic term and measured angular displacement, confirming the proposed model structure. (d) Regression errors across all test cases.

E. Model Calibration

To empirically identify the coefficients for our OC model, we developed a data collection procedure using the setup shown in Fig. 4(a). We tested a set of rigid steel rods with diameters (d) of 2.0, 3.0, 4.0, and 5.0 mm. For each rod, the gripper was oriented at various tilt angles (θ) ranging from 0 to 30 degrees relative to the object's axis. The gripper's linear actuators were then commanded to travel a specified number of steps (n), inducing a rotation. An external camera tracked a marker affixed to the distal tip of the object, and a vision algorithm computed the resulting angular displacement, Θ_{measured} , from the initial and final poses. This process yielded a dataset mapping the inputs (d, n, θ) to the measured rotational output.

The compensated OC model (Eq. 10) is linear with respect to the combined kinematic term. We performed the linear regression on the collected data. The regression yielded a strong linear fit with an R^2 value of 0.9949, validating the structure of our proposed model. The resulting slope and intercept directly provide the empirical coefficients, giving a gain coefficient of $K_{\text{OC}} = 0.53$ and an offset coefficient of $C_{\text{OC}} = -0.43$.

The experimental results, summarized in Fig. 4, provide further support for the model's validity. Fig. 4(c) shows a clear and strong linear relationship between the linear displacement of the fingers and the measured angle for all tested diameters, which is the core assumption of our rack-and-pinion analogy. Furthermore, the box plot in Fig. 4(d) indicates that the regression errors are consistently small, centered around zero, and maintain a low variance across the different object diameters. This demonstrates that the model generalizes well for objects within the OC state. The data in Fig. 4(b) provides additional qualitative validation, showing that achieving a constant 90° rotation requires greater linear displacement for both larger diameter objects and larger tilt angles, which aligns perfectly with the $1/d$ and $\cos(\theta)$ terms in our formulation.

V. SIMULATION ANALYSIS

To further evaluate the proposed framework, we conducted finite-element simulations using a simplified two-dimensional cross-sectional model. The setup consisted of two identical silicone pads with a thickness of 2.5 mm and a length of 20 mm, together with two metallic rods of diameters 0.2 mm and 0.5 mm. The silicone pads were modeled as linearly elastic materials with a Young's modulus of $E \approx 0.59$ MPa and a Poisson's ratio of $\nu \approx 0.49$, while the metallic rods were approximated as rigid bodies. The rods were fixed at their geometric centers, and external displacements were applied to the pads to capture compression and torsional interaction.

As illustrated in Fig. 5(a), the first simulation investigated the grasping process, in which the two silicone pads moved symmetrically toward the center until full contact was established with the rod. This analysis characterized the pressure distribution and the induced deformation during clamping, while inter-pad friction and viscoelastic effects

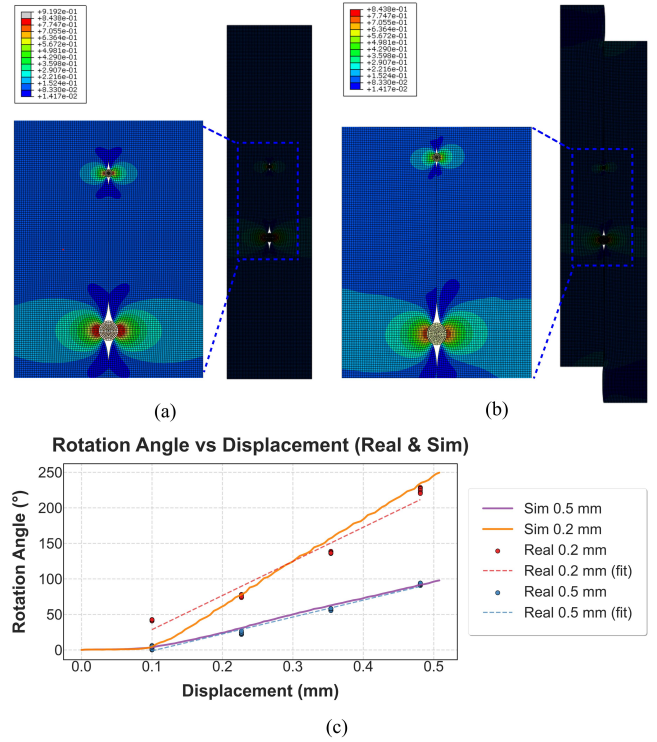


Fig. 5. **Simulation results.** (a) Grasping of rods with diameters 0.2 mm and 0.5 mm. (b) Twisting induced by reciprocal pad displacements. (c) Comparison between simulation and experimental results, showing close agreement for the 0.5 mm rod and minor deviations for the 0.2 mm rod.

were neglected for simplicity. Fig. 5(b) depicts the second simulation, where the silicone pads were driven with reciprocal vertical displacements. Through surface contact, this motion generated a torsional moment on the rod, which was constrained at its center. The resulting angular displacement of the rod was recorded as a function of pad displacement for both diameters (0.2 mm and 0.5 mm). To focus on the geometric relation between pad displacement and rod rotation, several simplifications were introduced: creep and hysteresis of the silicone were ignored, sliding friction at the contact interface was not considered, and the simulation was restricted to quasi-static loading.

As shown in Fig. 5(c), since this study fully reproduced the closed configuration regime, the results were fitted to the CC model in Eq. (12). The regression yielded $K_c = 0.53$, $A_c = -11.73$, and $C_c = -0.55$, which closely match the physical response of the metallic rods under torsional loading. Combining these simulation-derived coefficients for the CC regime with the empirically identified OC coefficients from Section IV-E, the final unified model is expressed as:

$$\Theta_{\text{deg}}(d, n, \theta) = \begin{cases} 0.53 \left(\frac{720sn}{\pi d} \cos(\theta) \right) - 0.43 & \text{OC,} \\ 0.53 \left(\frac{720sn}{\pi d} \right) - \frac{11.73}{d} - 0.55 & \text{CC.} \end{cases} \quad (13)$$

For the case of a 0.5 mm diameter rod, the simulated curve in Fig. 5(c) closely matches the experimental measurements, demonstrating the accuracy of the proposed model. In contrast, for the 0.2 mm rod, noticeable discrepancies arise. This

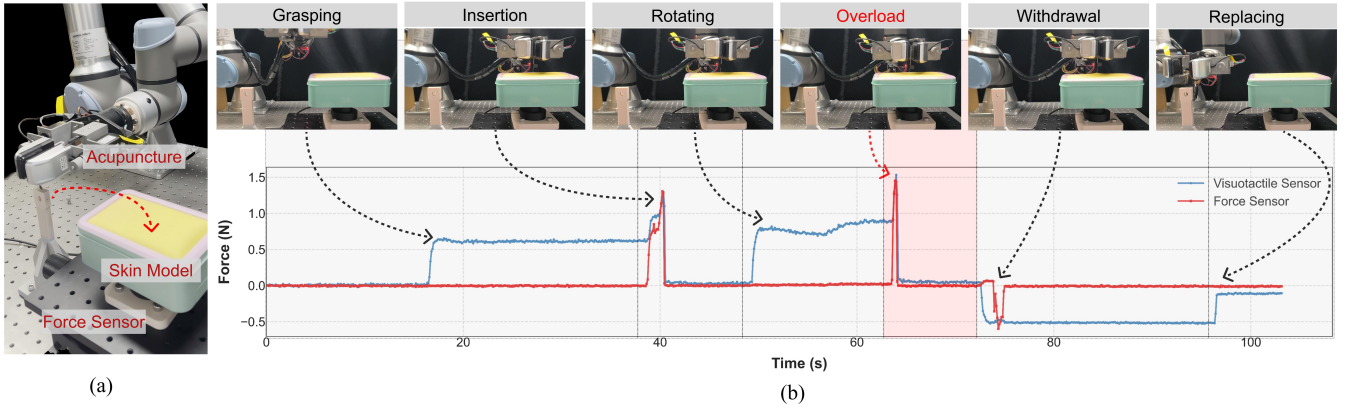


Fig. 6. **Acupuncture experiment setup and results.** (a) The VAIT gripper manipulates an acupuncture needle into a skin model positioned over a force sensor. (b) Time-series data comparing the force estimated by the visuotactile sensor (blue) with the force sensor (red). The plot shows phases of grasping, insertion, twisting, a deliberate overload event triggering the safety release, withdrawal, and replacement.

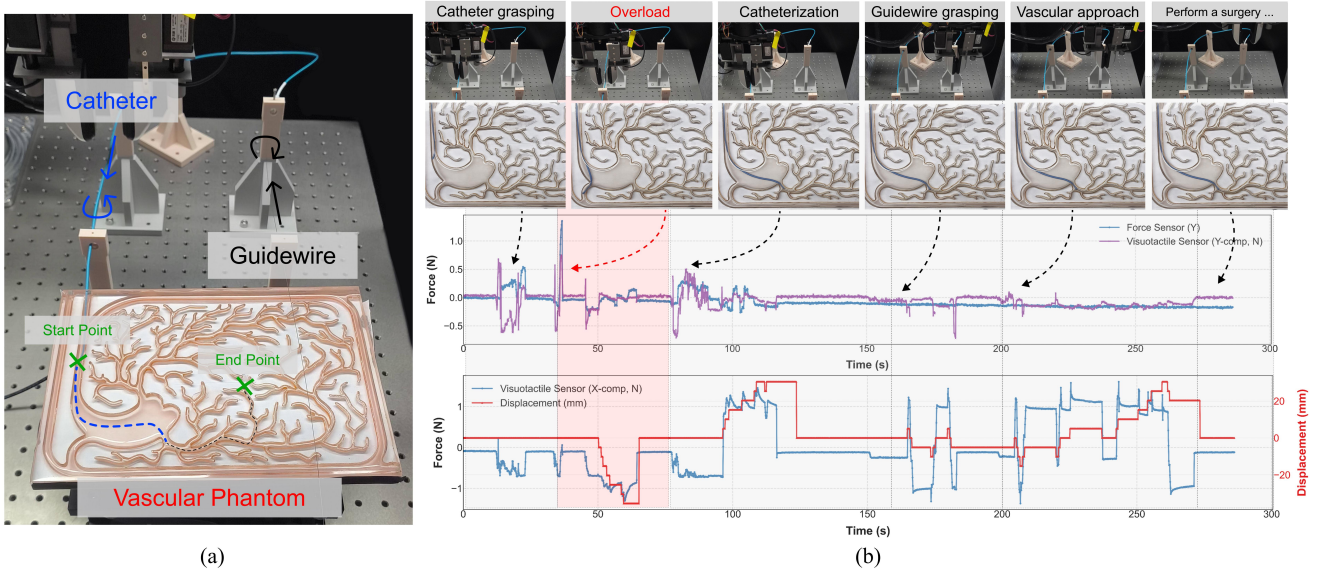


Fig. 7. **Vascular intervention experiment setup and results.** (a) The VAIT gripper manipulates a catheter and a guidewire through a vascular phantom. (b) Top: Time-series data for push/pull forces (Y-component) comparing the visuotactile sensor (purple) with a force sensor (blue). Bottom: Time-series data plotting twisting shear force (X-component, blue) against motor displacement (red), demonstrating the state switch from OC to CC.

is attributed to complex contact phenomena inherent to soft viscoelastic materials that were not captured in our simplified quasi-static model, such as time-dependent viscoelastic creep and other non-linear frictional effects that are challenging to model accurately [17]. Nevertheless, the simulation indicates that these errors diminish once the imposed displacement exceeds 0.2mm. Based on this observation, the minimum motor step in subsequent experiments was set to be larger than 0.2mm in order to mitigate the influence of these unmodeled factors.

VI. EXPERIMENTS

To validate the capabilities of the VAIT framework, we designed two complementary experiments performed via teleoperation with system states recorded at 10 Hz. The first experiment, an acupuncture simulation, focuses on evaluating

the core manipulation primitive: high-precision twisting of a single, semi-rigid, ultra-thin object. It serves to demonstrate the system's fundamental reliability and its real-time visuotactile force tracking for safety monitoring. The second experiment, a simulated interventional surgery, evaluates the system's versatility and its adaptive, piecewise control model. This is achieved by requiring the manipulation of multiple, highly flexible DLOs of varying diameters, necessitating a transition between the Open and Closed operational regimes.

A. Experiment 1: Acupuncture

This experiment tests the system's core capabilities in a task requiring fine dexterity and force sensitivity. The setup (Fig. 6(a)) consists of the VAIT end effector, an acupuncture needle, and a skin tissue phantom positioned over a force sensor. The teleoperated procedure involved six sequential

actions: (1) grasping the needle from a holder, (2) inserting it into the tissue model, (3) performing in-hand twisting manipulations, (4) deliberately overloading the system against the hard underlying material in the tissue phantom to test the safety release, (5) withdrawing the needle, and (6) returning it to the holder.

The time-series data in Fig. 6(b) demonstrates three key outcomes. First, the successful execution of the entire sequence validates the system’s ability to perform reliable grasping and twisting of ultra-thin objects. Second, the force estimated from the visuotactile sensor (blue line) closely follows the real forces (red line), confirming the effectiveness of our real-time force tracking, especially in capturing the distinct spikes during insertion and withdrawal. Third, the plot validates the automatic safety mechanism, showing the force dropping to zero as the gripper releases upon exceeding the 1 N threshold during both accidental (approx. 40s) and deliberate (approx. 65s) overloads. The consistent positive shear reading upon grasping is attributed to the preloading of the soft sensor pads, while the negative reading after withdrawal is likely due to material hysteresis.

B. Experiment 2: Vascular Intervention

This experiment evaluates the system’s versatility and adaptive control by simulating a guidewire-in-catheter procedure on highly flexible DLOs of varying diameters (a 2 mm catheter and a 0.3 mm guidewire). The task, performed in the vascular phantom shown in Fig. 7(a), required a six-stage sequence: (1) grasping the catheter, (2) deliberately navigating into a wrong branch to trigger the safety overload, (3) retracting and performing correct catheterization using twisting, (4) releasing the catheter and grasping the guidewire, (5) navigating the guidewire using push/pull and twisting motions, and (6) delivering the guidewire tip to the target end point.

1) Analysis of Push/Pull Forces: The top graph in Fig. 7(b) shows the time-series data for the push/pull forces. The estimated force closely tracks real forces throughout the operation, and the safety mechanism is again shown to activate successfully during the overload phase. An initial deviation upon first grasping the catheter is likely due to the complex 3D material settling of the soft pads, which is not fully captured by the force sensor until the grasp stabilizes.

2) Analysis of Twisting Forces and Regime Switching: The bottom graph in Fig. 7(b) provides a clear validation of our adaptive, piecewise control model. During catheter manipulation (approx. 50-140s), the system operates in the OC state, where the measured shear force is directly proportional to the commanded motor displacement. When switching to the ultra-thin guidewire (approx. 150s onwards), the system enters the CC state. In this high-contact state, the relationship becomes fundamentally different; very small motor displacements generate significant, amplified shear forces as the compressed pads slide against each other. This result validates the necessity of our dual-regime model for adaptively handling objects of drastically different scales.

VII. CONCLUSION

In this work, we presented VAIT, a framework that addresses the challenge of precise, in-hand twisting of ultra-thin slender objects. By co-designing a multi-DoF gripper with integrated visuotactile sensing, our system bridges the gap between the precision of specialized hardware and the versatility of general-purpose hands. The core of our approach is an adaptive, piecewise control model that distinguishes between a kinematically driven OC state and a deformation-dominated CC state. This allows a single end-effector to robustly handle objects across a wide range of diameters, from 5 mm rods down to submillimeter filaments. We demonstrated how shear force information, derived from the visuotactile sensors, enables real-time force tracking and the implementation of a critical safety release mechanism. Experimental results from simulated acupuncture and surgical tasks validated the framework’s reliability in twisting ultra-thin objects, its versatility in handling multiple flexible DLOs, and the necessity of its adaptive, dual-regime model.

While our results are promising, we recognize several limitations. The current pose control model is feed-forward; it calculates the necessary motor steps to achieve a target angle but does not actively correct for rotational errors during execution. Furthermore, our demonstrations were performed via teleoperation, which showcases the system’s capabilities but does not yet constitute full autonomy.

Future work will focus on three primary directions. First, we plan to develop a closed-loop controller that uses the real-time angle and force feedback from the visuotactile sensors to achieve more robust and accurate twisting performance. Second, we will integrate the VAIT manipulation primitive into a higher-level task planner, enabling the system to autonomously perform complex, multistage tasks like surgical suturing or cable connector insertion. Finally, the rich data stream from the visuotactile sensor presents an opportunity to explore learning-based approaches to refine the control model, allowing it to adapt to novel objects and more complex contact dynamics online.

APPENDIX

Symbol	Description
d	Diameter of slender object (mm)
n, s	Motor step count and linear travel per step ($s = 0.254$ mm/step)
θ	Object orientation (deg); $\theta_{\text{norm}}^\circ, \theta_{\text{raw}}^\circ$ normalized/unsigned forms
$\bar{\theta}_t, \theta_t$	Rolling-averaged and smoothed orientation estimates; n_t window size, α smoothing factor
$(\Delta x, \Delta y)$	Centerline endpoint displacement (image domain)
Θ_{ideal}	Ideal angular displacement (deg)
Θ_{measured}	Measured angular displacement (deg)
$\Theta_{\text{OC}}, \Theta_{\text{CC}}, \Theta_{\text{deg}}$	Predicted angles in Open/Closed/Unified models
$m_i, (d_{x,i}, d_{y,i}), N$	Optical-flow magnitude, direction vector, number of markers
S_x, S_y, S	Shear integrals along x, y ; S resultant shear integral
F_x, F_y	External force components (N)
a, b	Mapping coefficients ($a = 0.0053, b = 1.0788$)
$K_{\text{OC}}, C_{\text{OC}}$	Open Configuration model coefficients ($K_{\text{OC}} = 0.53, C_{\text{OC}} = -0.43$)
$K_{\text{CC}}, A_{\text{CC}}, C_{\text{CC}}$	Closed Configuration model coefficients ($K_{\text{CC}} = 0.53, A_{\text{CC}} = -11.73, C_{\text{CC}} = -0.55$)
E, ν	Silicone Young's modulus ($E \approx 0.59$ MPa) and Poisson's ratio ($\nu \approx 0.49$)
Abbrev.	Meaning
VAIT	Visuotactile-guided Adaptive In-hand Twisting
DoF	Degree(s) of Freedom
DLO	Deformable Linear Object
OC, CC	Open Configuration, Closed Configuration
VS, LG, LM, GM	Visuotactile Sensor, Linear Guide, Linear Motor, Gripper Motor

REFERENCES

- [1] Y. She, S. Wang, S. Dong, N. Sunil, A. Rodriguez, and E. Adelson, "Cable manipulation with a tactile-reactive gripper," *The International Journal of Robotics Research*, vol. 40, no. 12-14, pp. 1385–1401, 2021.
- [2] M. Mizukami and M. Makihara, "Fiber-handling robot and optical connection mechanisms for automatic cross-connection of multiple optical connectors," *Optical Engineering*, vol. 52, no. 7, pp. 076116–076116, 2013.
- [3] X. Bao, S. Guo, C. Yang, and L. Zheng, "Haptic interface with force and torque feedback for robot-assisted endovascular catheterization," *IEEE/ASME Transactions on Mechatronics*, vol. 29, no. 2, pp. 1111–1125, 2024.
- [4] X. Bao, S. Guo, Y. Guo, C. Yang, L. Shi, Y. Li, and Y. Jiang, "Multilevel operation strategy of a vascular interventional robot system for surgical safety in teleoperation," *IEEE Transactions on Robotics*, vol. 38, no. 4, pp. 2238–2250, 2022.
- [5] M. Yu, K. Lv, C. Wang, Y. Jiang, M. Tomizuka, and X. Li, "Generalizable whole-body global manipulation of deformable linear objects by dual-arm robot in 3-d constrained environments," *The International Journal of Robotics Research*, vol. 44, no. 4, pp. 607–639, 2025.
- [6] W. Yuan, S. Dong, and E. H. Adelson, "GelSight: High-resolution robot tactile sensors for estimating geometry and force," *Sensors*, vol. 17, no. 12, p. 2762, 2017.
- [7] Y. Yan, S. Guo, C. Lyu, D. Zhao, and Z. Lin, "SEA-based humanoid finger-functional parallel gripper with two actuators: PG2 gripper," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–13, 2022.
- [8] G. Li, R. Wang, P. Xu, Q. Ye, and J. Chen, "The developments and challenges towards dexterous and embodied robotic manipulation: A survey," *arXiv preprint arXiv:2507.11840*, 2025.
- [9] A. I. Weinberg, A. Shirizly, O. Azulay, and A. Sintov, "Survey of learning-based approaches for robotic in-hand manipulation," *Frontiers in Robotics and AI*, vol. 11, p. 1455431, 2024.
- [10] M. Yu, Y. Jiang, C. Chen, Y. Jia, and X. Li, "Robotic in-hand manipulation for large-range precise object movement: The rgmc champion solution," *arXiv preprint arXiv:2502.07472*, 2025.
- [11] G. Xie, R. Holladay, L. Chin, and D. Rus, "In-hand manipulation with a simple belted parallel-jaw gripper," *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1334–1341, 2024.
- [12] G. He, C. Sparks, and N. Gravish, "Grasping and rolling in-plane manipulation using deployable tape spring appendages," *Science Advances*, vol. 11, no. 15, p. eadt5905, 2025.
- [13] O. Azulay, K. Kondap, J. Drake, S. Xie, H. Li, S. Chitta, and K. Goldberg, "Adaptive cable manipulation around constrained fixtures with bi-manual slack control," in *arXiv preprint*, 2025.
- [14] M. Yu, B. Liang, X. Zhang, X. Zhu, X. Li, and M. Tomizuka, "In-hand following of deformable linear objects using dexterous fingers with tactile sensing," in *arXiv preprint arXiv:2403.12676*, 2024.
- [15] S. Soutukorva, J. Saukkoriipi, and M. Suomalainen, "Gripper fingers for in-hand cable rotation with a parallel gripper," in *IEEE International Conference on Robotics and Automation (ICRA) Workshop*, 2025.
- [16] S. Li, Z. Wang, C. Wu, X. Li, S. Luo, B. Fang, F. Sun, X.-P. Zhang, and W. Ding, "When vision meets touch: A contemporary review for visuotactile sensors from the signal processing perspective," *IEEE Journal of Selected Topics in Signal Processing*, vol. 18, no. 3, pp. 267–287, 2024.
- [17] D. Wang, G. de Boer, A. Neville, and A. Ghanbarzadeh, "A review on modelling of viscoelastic contact problems," *Lubricants*, vol. 10, no. 12, p. 358, 2022.